Task 1: A Profile Header

```
import Foundation

struct ProfileHeader {
    let name: String
    let description: String
    let posts: Int
    let followers: Int
    let following: Int

    func showStatistic(_ title: String, _ value: Int) {
        print("\(title): \(value)")
    }

    func display() {
        print("========== PROFILE ==========")
        print("Banner height: 160")
        print("Avatar: 👤 ")
        print("Name: \(name)")
        print("Description: \(description)")
        print("----------------------------")
        showStatistic("Posts", posts)
        showStatistic("Followers", followers)
        showStatistic("Following", following)
```

```
        print("============================")
    }
}

let profile = ProfileHeader(
    name: "Manoj",
    description: "Swift Developer",
    posts: 12,
    followers: 250,
    following: 180
)

profile.display()
```

Output

```
========== PROFILE ==========
Banner height: 160
Avatar: 👤
Name: Manoj
Description: Swift Developer
----------------------------
Posts: 12
Followers: 250
Following: 180
============================
```

Task 2: A Responsive Product Grid

```
import Foundation

struct Product {
    let name: String
    let icon: String
}

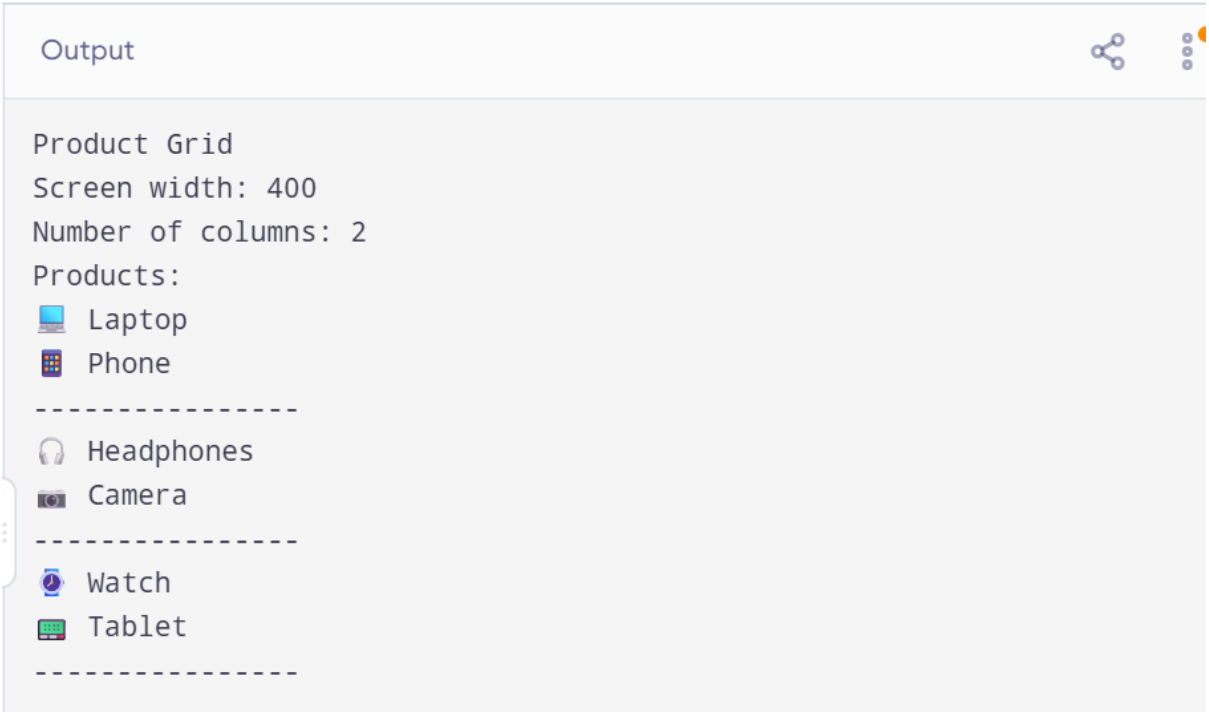
let products = [
    Product(name: "Laptop", icon: "💻"),
    Product(name: "Phone", icon: "📱"),
    Product(name: "Headphones", icon: "🎧"),
    Product(name: "Camera", icon: "📷"),
    Product(name: "Watch", icon: "⌚"),
    Product(name: "Tablet", icon: "📟")
]

let screenWidth = 400
let minimumCardWidth = 140
let columns = max(1, screenWidth / minimumCardWidth)

print("Product Grid")
print("Screen width: \(screenWidth)")
print("Number of columns: \(columns)")
print("Products:")
```

```
for (index, product) in products.enumerated() {
    print("\(product.icon) \(product.name)")

    if (index + 1) % columns == 0 {
        print("----------------")
    }
}
```



Task 3: How Sizing Really Works

```
import Foundation
let screenWidth = 360.0
let textWidth = 200.0
let yellowBackgroundWidth = screenWidth
```

```
print("A. Text with frame before background")
print("Text width: \(Int(textWidth))")
print("Background width: \(Int(textWidth))")
print("\nB. Background before frame")
print("Text uses its natural width")
print("Background frame width: \(Int(textWidth))")
let leftWidth = screenWidth / 3
let rightWidth = screenWidth - leftWidth
print("\nC. Two-colour bar")
print("Left section: \(Int(leftWidth))")
print("Right section: \(Int(rightWidth))")
print("Total width: \(Int(screenWidth))")
```

Output

```
A. Text with frame before background
Text width: 200
Background width: 200

B. Background before frame
Text uses its natural width
Background frame width: 200

C. Two-colour bar
Left section: 120
Right section: 240
Total width: 360
```